KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2020

EE 2113

(Electrical Machines)

Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any Two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

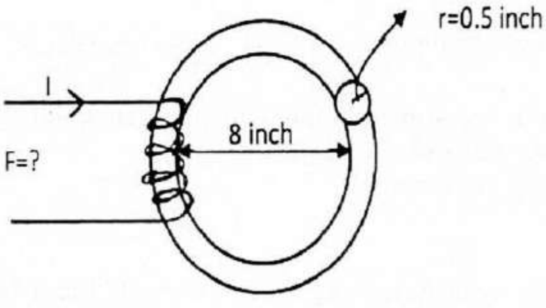
iii) Assume reasonable data if any missing.

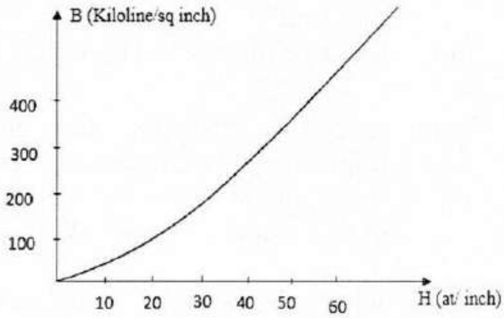
SECTION – A

1(a). Does a magnetic circuit preserve analogous relationship like an electrical circuit? Justify your answer with two suitable examples. 12

1(b). What is hysteresis loop? Draw a hysteresis loop with coercive force, residual magnetism and retentivity properly identified. 08

1(c). Find the value of mmf for the magnetic circuit shown in figure below, Given $\phi = 0.0008Wb$. 10





2(a). Mathematically explain how a revolving field is produced in an induction motor if a 3-φ balanced supply is connected to its 3-φ windings. Consider only the cases θ=0° and θ=60°. 15

2(b). A 500V, 3 Phase, 8 Pole Y-connected induction motor has the following equivalent circuit parameters: $R_1 = 0.13\Omega$, $R_2 = 0.32\Omega$, $X_1 = 0.6\Omega$, $X_2 = 1.48\Omega$, $X_M = 34.6\Omega$. Determine: 15

i) Internal power and torque for S = 0.03,

ii) The maximum torque developed, and

iii) The starting torque.

3(a). What are the factors affecting alternator size? How armature reaction affects the terminal voltage of alternators? Is there any dependency of armature reaction on the types of load connected to the alternators? Justify. 18

3(b). Explain load sharing of alternators in terms of change of excitation. 12

SECTION – B

4(a).	Mention the essential parts of a DC generator and write down the functions of commutator, brush and pole shoes.	10
4(b).	With proper circuit diagrams classify DC generators according to the way in which their fields are excited. Draw i) O.C.C of a series wound DC Generator ii) External characteristic of DC shunt generator.	12
4(c).	Define critical resistance. The coil of a four-pole generator is rotating at a speed of 1200 RPM. The flux per pole is $2 \times 10^{-5} Wb$. Find the voltage induced in the coil if it has 100 turns.	08
5(a).	What is back emf? Write down the voltage equation of a DC motor and show the significance of back emf.	10
5(b).	Describe the method by which speed of DC motor can be controlled above rated speed.	10
5(c).	A 25-kW, 250V, DC shunt generator has armature and field resistances of 0.06 Ω and 100Ω respectively. Determine the total armature power developed when i) working as a generator delivering 25 kW output. ii) as a motor taking 25 kW input.	10
6(a).	What are the losses of transformer? How can these losses be determined? Describe the open circuit test of transformer.	12
6(b).	Derive the condition for maximum efficiency of a transformer.	10
6(c).	A 25kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000V, 50 Hz supply. Determine the value of maximum flux interact with the secondary winding if leakage drops in the primary winding is 30V. Neglect no-load primary current.	08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2020

Math 2113

(Linear Algebra and Vector Analysis)

Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any TWO questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

1(a). Discuss briefly with example about the arithmetic operations and rules in matrices which are different from that operations and rules in real numbers. 10

1(b). Reduce the following matrix to its echelon form, canonical form and finally normal form sequentially: 20

$$\begin{bmatrix} 0 & p & 1 & -2 \\ 1 & q & 1 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix}. \text{ Hence find its rank.}$$

Where p and q are the last two digits of your class roll number measured from the rightmost sequentially. [Hint: Roll: 1813001, then $p = 1, q = 0$, Roll: 1813023, then $p = 3, q = 2$].

2(a). Determine the values of k such that the following system of linear equation has (i) a unique solution, (ii) more than unique solution and (iii) no solution 22

$$\begin{cases} kx + y + z = 1 \\ x + ky + z = 1 \\ x + y + kz = 1 \end{cases}$$

Also find the solutions of exist.

2(b). Define row space and null space with example. 08

3(a). Find the eigen values and eigen vector correspond to the largest eigen value of A, if exist ,where $A = \begin{bmatrix} 2 & (p+1) \\ (1+q) & 1 \end{bmatrix}$, where p and q are the last two digit of your class roll number measured from the rightmost sequencially.[Hint: Roll: 1813001, then $p = 1, q = 0$, Roll: 181023, then $p = 3,\ q = 2$]. 15

3(b). Let $X = [1 \quad p \quad q]^t$, $Y = [1 \quad 2 \quad 3]^t$, $Z = [1 \quad 1 \quad 2]^t$. Are they formed the vector space $\mathbb{R}^3$[$i.e$ are they linearly independent vectors of euclidean 3-vectors space $\mathbb{R}^3$]? If not, then express the dependent vector as a linear combination of independent vectors of $\mathbb{R}^2$.Where p and q are the last two digits of your class roll number measured from the rightmast sequencially.[Hint: Roll: 1813001, then $p = 1,\ q = 0$, Roll: 1813023, then $p = 3, q = 2$]. 15

SECTION – B

4(a). Sketch the space curve whose position vector is $(R\cos t\,\hat{\imath} + R\sin t\,\hat{\jmath})$. Hence find unit tangent vector $\widehat{T}$, Normal vector $\widehat{N}$ and Binomial vector $\widehat{B}$. Where R is the sum of the last two digits of your class roll number. 15
[Hints: Roll 1813005 then $R = 0 + 5 = 5$; Roll : 1813027 then $R = 2 + 7 = 9$].

4(b). Test the nature of the vector point function $\frac{G\,\overline{r}}{r^2}$ (by using vector operator '∇'), where G is constant and $\overline{r}$ is the position vector. Hence if possible, find the scalar potential function $\emptyset$ such that $\emptyset(R, 0, R)$=1, where R is the sum of the last two digits of your class roll number. 15
[Hints: Roll 1813005 then $R = 0 + 5 = 5$; Roll: 1813027 then $R = 2 + 7 = 9$].

5(a). Define line integral. Find the work done for moving a particle in the force field 15
$\overline{F} = 3x^2\hat{\imath} + (2xz - y)\hat{\jmath} + z\hat{k}$ along
(i) the straight line from (0,0,0) to (2,1,3)
(ii) the curve defined by $x^2 = 4y,\ 3x^3 = 8z\ from\ x = 0\ to\ x = 2$.

5(b). Find equations for the tangent plane and normal line to the surface $z = x^2+y^2$ at the point $(2, -1, 5)$. 15

6(a). Evaluate $\oint(3x^2 + 2y)dx - (x + 3\cos y)dy$ around the parallelogram having vertices at $(0,0), (2,0), (3,1), and\ (1,1)$. 10

6(b). Verify the divergence theorem for $\overline{A} = 2x^2y\hat{\imath} - y^2\hat{\jmath} + 4xz^2\hat{k}$ taken over the region in the first octant bounded by $y^2 + z^2 = 9$ and $x = 2$. 20

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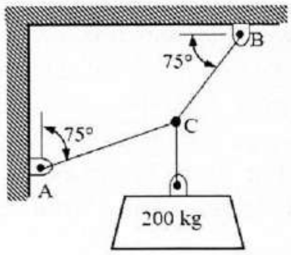
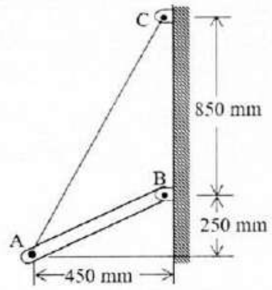
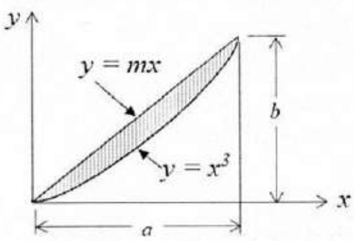
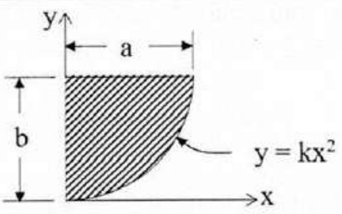
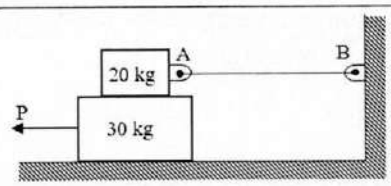
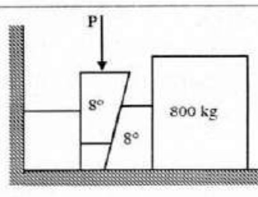
KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2020
ME 2113
(Statics and Solid Mechanics)

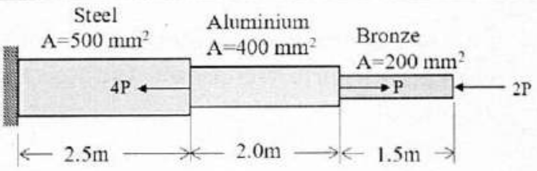
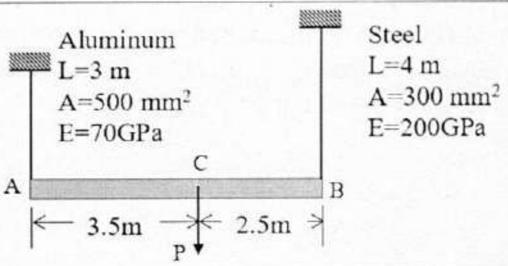
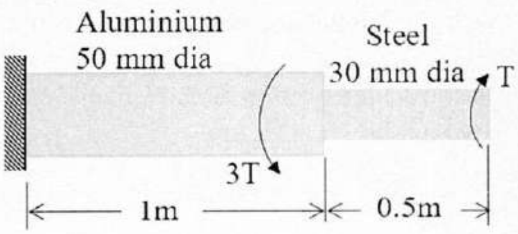
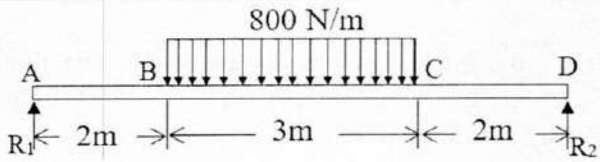
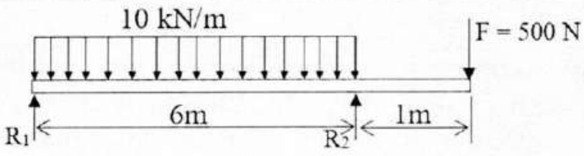
Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a).	Two cables are tied together at C and are loaded as shown in figure. Determine the tension in cable AC and BC.		14
1(b).	Rod AB is held in place by the cord AC as shown in figure. The moment about B of the force exerted by the cord at A is 1000 N-m. Determine the tension in the cord.		16
2(a).	Distinguish between centroid and center of gragity.		04
2(b).	By direct integration, determine the centroid of the area as shown in figure.		12
2(c).	By direct integration, determine the moment of inertia of the shaded area with respect to the x axis.		14
3(a).	The coefficients of friction are $\mu_s = 0.40$ and $\mu_k = 0.30$ between all surfaces of contact. Determine the smallest force P required to start the 30 kg block moving if cable AB (i) is attached as shown, (ii) is removed.		15
3(b).	Two 8° wedges of negligible weight are used to move and position the 800 kg block. Knowing that the coefficient of static friction is 0.25 at all surfaces of contact, determine the smallest force P that should be applied as shown to one of the wedges.		15

SECTION – B

4(a).	An aluminum rod is rigidly attached between a steel rod and a bronze rod as shown in figure. Axial loads are applied at the position indicated. Find the maximum value of P that will not exceed a stress in steel of 140 MPa, in aluminum of 90 MPa or in bronze of 100 MPa.		15
4(b).	The rigid bar AB attached to two vertical rods as shown, is horizontal before the load P is applied. Determine the vertical movement of P if its magnitude is 50 kN.		15
5(a).	A compound shaft consisting of a steel segment and an aluminium segment is acted upon by two torques as shown in figure. Determine the maximum permissible value of T if the angle of twist is restricted to 5°. For steel, G = 80 GPa and for aluminium, G = 30 GPa.		15
5(b).	Find the mid-span value of the deflection for the beam loaded as shown in figure. Use $E = 10\ GPa$ and $I = 20 \times 10^6\ mm^4$.		15
6(a).	Draw the shear force diagram and bending moment diagram of the beam loaded as shown in figure.		15
6(b).	A cantilever beam 50 mm wide by 100 mm high and 6 m long, carries a load that varies uniformly zero at the free end to 1 kN/m at the wall. Compute the magnitude and location of the maximum flexural stress.		15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2020
ME 2115
(Fluid Mechanics)

Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any TWO questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). State and explain Newton's law of viscosity. What is the effect of temperature on viscosity of water and that of air? 12

1(b). Define surface tension and vapor pressure of fluids. 06

1(c). Calculate the loss of power due to friction for the following bearing. Consider the shaft speed is 210 rpm and viscosity of the oil is 0.75 $N.s/m^2$. 12

2(a). Show that the centre of pressure is below the centre of gravity of an inclined submersed object. 12

2(b). What are the applications of U-tube and inverted U-tube differential manometers? 04

2(c). In a typical salt gradient solar pond, the density of water increases in the gradient zone and it can be expressed as 14

$$\rho = \rho_0 \sqrt{(1 + tan^2(\frac{\pi}{4}\frac{s}{H})}$$

Where ρ_0 is the density on the water surface, s is the vertical distance measured downward from the top of the gradient zone, i.e $s = -2$, and H is the thickness of the gradient zone. For $H = 5m$, $\rho_0 = 1050\ kg/m^3$, and a thickness of 0.8m for the surface zone, calculate the absolute pressure at the bottom of the gradient zone. Note that solar pond is divided into three zones from the top surface zone, gradient zone and storage zone.

3(a). Describe the followings with neat sketch and proper condition: 06
i) Stable equilibrium
ii) Neutral equilibrium
iii) Unstable equilibrium

3(b). What is meant by metacentre and metacentric height? 06

3(c). Explain the concept of streamline and streamtube with neat sketch. 06

3(d). Derive the three dimensional continuity equation from first principle. 12

SECTION – B

4(a). What are the difference between 'Lagrangian Method' and 'Eulerian Method'? 05

4(b). Derive the Bernoulli's equation from the Fundamental equation. What are the limitations of this equation? 15

4(c). A 25cm diameter pipe carries oil of specific gravity 0.9 at a velocity of 3m/s. At another section, the diameter is 20cm. Find the velocity at this section and also mass flow rate of flow of oil. 10

5(a). What is 'Energy Grade line' and 'Hydraulic Grade line'? Draw the energy grade line and hydraulic grade line for a flow through variable area. 12

5(b). Derive the integral equation of the mass conservation equation using Reynolds transport theorem. 12

5(c). Explain the followings with neat sketch : 06

i) Pressure drag ii) Friction drag

6(a). For the constant pressure fluid flow over a flat plate, the integral boundary layer equation is given as : 18

$$\rho \frac{d}{dx}\int_0^{\delta}(v-u)u\,dy = \mu \frac{\delta u}{\delta y}\bigg|_{y=0} \quad .$$

Using the velocity profile $\frac{u}{v} = \frac{3}{2}\frac{y}{\delta} - \frac{1}{2}(\frac{y}{\delta})^3$, show that the boundary layer thickness is $\frac{\delta}{x} = \frac{4.64}{Re_{x^{1/2}}}$.

Also show the skin fiction coefficient is $c_f = \frac{0.5774}{Re_{x^{1/2}}}$.

6(b). Given that a laminar boundary layer at zero pressure gradient over a flat plate is described by the velocity profile: $\frac{u}{v} = \sin(\frac{\pi}{2}\frac{y}{\delta})$. Determine the displacement thickness and the momentum thickness. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2020

CSE 2113

(Computer Programming)

Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any Two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

1(a). What are the differences between Compilers and Pseudo-compliers? Explain Octave order of precedence by describing the execution of following code written in Octave. 12

```
x=5;
y=4;
(x+y)^2-(x+y)/(x-y)^2+sqrt((x+y)/(2*x-y))
```

1(b). The distance traveled by a ball falling in the air is given by equation 10

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

Use MATLAB to calculate the position of the ball at time $t = 5s$ if $x_0 = 10m$, $v_0 = 12ms^{-1}$, and $a = -9.81m/s^2$.

1(c) Explain different formats of displaying a number in Octave with example. 08

2(a). What are the differences between symbolic and numerical calculation? 05

2(b). Explain with example when will you use the followings: 15

i) for loop; ii) while loop; iii) if-elseif-else-end structure.

2(c). Design a Octave program that reads an imput temperature in degrees Fahrenheit, converts it to an absolute temperature in Kelvin and write out the result and finally break. 10

3(a). Consider the following system of linear equations as given below: 18

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 = b_3$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 = b_4$$

Write an Octave code to solve the system of linear equations using gauss elimination method.

3(b). A Matrix 'M' is given by – 12

$$M = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ -1 & 3 & 2 \end{bmatrix}$$

By using Octave commands, find the eigen values and eigen vectors in a single statement.

SECTION – B

4(a). Suppose $A = \begin{bmatrix} 2 & 4 & 6 \\ 4 & 7 & 9 \\ 8 & 1 & 3 \end{bmatrix}$ 15

Write down the output of the following code written in octave.

```
i) a = find (A>5)
ii) a1= A(find(A>5))
iii)[b, d] = find( A>5)
iv) e = A-diag (diag(A))
v) f = e+5
vi) j = fliplr(f)
vii) k= flipud(f)
viii) h= f(:,length (f):-1:1)
ix) m = sum(sum(f))
x) n = m+sum(sum(A))
```

4(b). Write down the function of the following command with example: 10
i) mesh grid ii) magic iii) sort iv) rand v) mean

4(c). A code is given as: 05

```
spmd= 1:2:23
M = reshape (v,3,4)
M(2,: )= [ ]
M(:,3 )= [ ]
return = ones (size(M))
```

Is there any error in the code? Write down the output of the code.

5(a). Write a MATLAB program to evaluate a function $f(x, y)$ for any two user-specified values x and y. 10
The function $f(x, y)$ is given as :

$$f(x,y) = \begin{cases} x + y & x \geq 0 \; and \; y \geq 0 \\ x + y^2 & x \geq 0 \; and \; y < 0 \\ x^2 + y & x < 0 \; and \; y \geq 0 \\ x^2 + y^2 & x < 0 \; and \; y < 0 \end{cases}$$

5(b). How can you plot multiple graph in the same plot? Explain with example. What are the differences between these methods? 10

5(c). A parametric equation is given as: 10
$x = r\cos\theta, y = r\sin\theta,\ z = 4r$ with $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 2$. Write an Octave code to plot the surface created by the above equation and show xy, yz and zx projection in same figure window.

6(a). Consider the following data: 10

x	0	0.6	0.9	1.16	1.18	1.19	1.24	1.48	1.92	3.12	4.14	5.34	6.22
y	0	1.2	2.4	3.6	4.8	6.0	7.2	8.4	9.6	10.8	12.0	13.2	14.4

i) Among power, exponential, logarithmic and reciprocal function which one best fits the above data? Give your explanation.

ii) Write an Octave code for fitting the data with 5th order polynomial.

6(b). Briefly explain the following function with example: 20

i) polyval ii) roots iii) conv iv) deconv v) polyder

……End……